ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO3 SCENARIO-BASED ASSIGNMENT

Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

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Stage 1 – Interpret the Application Requirements

The proposed optoelectronic system combines biomedical sensing, automotive LiDAR, image sensing, UV detection, face recognition, photovoltaic power generation and advanced displays. It needs high responsivity, high sensitivity, low noise, sufficient bandwidth, good linearity, low power consumption and reliable integration.

- High responsivity converts received optical power into a useful photocurrent.
- High sensitivity and low NEP help detect weak optical signals.
- High bandwidth and short response time are important for LiDAR and fast imaging.
- CMOS/CCD sensors must provide suitable image quality, speed and power consumption.
- Flexible PV can provide auxiliary power in wearable applications.
- OLED, MicroLED and quantum-dot displays are selected based on brightness, efficiency, lifetime and flexibility.
- The complete system must be reliable, compact and thermally stable.

Application Requirement Statement: The system should accurately detect weak optical signals, provide high-speed LiDAR detection, support biomedical and imaging applications, use power efficiently and integrate suitable PV and display technologies.

Stage 2 – Photodetector Configuration

1. Responsivity

Responsivity is the parameter that converts received optical power into photocurrent.

$R = I_p / P$, therefore $I_p = R \times P$

2. PPG photocurrent variation

Given: $R = 0.60$ A/W and $\Delta P = 25\ \mu W = 25 \times 10^{-6}$ W

$\Delta I = R \times \Delta P = 0.60 \times 25 \times 10^{-6} = 15 \times 10^{-6}$ A

Answer: $\Delta I = 15\ \mu A$. PPG detects changes in reflected light caused by changes in blood volume. The signal can be processed to estimate pulse rate.

3. PIN and APD comparison

Parameter	PIN photodiode	APD photodiode

Internal gain	No internal multiplication	Avalanche multiplication provides gain
Sensitivity	Good for normal light levels	Very high for weak signals
Noise	Generally lower	Higher because of avalanche excess noise
Bias voltage	Lower	Higher
Complexity	Simple and easy to use	More complex bias and control
Applications	PPG and general sensing	LiDAR and low-light detection

4. LiDAR photocurrent

Given: $P = 3\ \mu W = 3 \times 10^{-6}$ W; $R = 0.80$ A/W

$I_p = R \times P = 0.80 \times 3 \times 10^{-6} = 2.4 \times 10^{-6}$ A

Answer: LiDAR photocurrent = 2.4 μA.

5. Other detector parameters

APD gain: Avalanche multiplication increases photocurrent and helps detect weak signals, but also increases noise and requires a higher bias voltage.

Dark current: Current flowing without incident light; it adds noise and reduces weak-signal accuracy.

Quantum efficiency (QE): QE = (collected electrons / incident photons) × 100%. Higher QE means a greater fraction of incident photons produces collected charge.

Noise Equivalent Power (NEP): The optical input power that gives a signal equal to the noise in a specified bandwidth. Lower NEP means better sensitivity.

Detectivity: A measure of weak-signal detection ability. Higher detectivity is better.

Linearity: Photocurrent should be proportional to optical power over the operating range. It improves measurement accuracy.

Design Decision 1: I recommend a PIN photodiode for PPG and an APD for LiDAR and low-light detection because PIN devices have low noise and simple operation, while APDs provide internal gain and high sensitivity.

Stage 3 – Noise and SNR Strategy (15 Marks)

SNR compares signal power with noise power.

Given: Signal = −22 dBm; noise = −37 dBm

SNR(dB) = Signal power (dBm) − Noise power (dBm) = −22 − (−37) = 15 dB

Linear SNR = 10^(15/10) = 31.62. Answer: SNR = 15 dB, or about 31.62 as a linear power ratio.

Shot noise occurs due to the discrete nature of electric charge. Dark-current noise is caused by current flowing in darkness. Avalanche noise is caused by the random multiplication process in an APD.

At low optical power, shot noise and dark-current noise can become important. APDs also have avalanche excess noise. The dominant source depends on detector parameters and operating conditions.

If optical power decreases while dark current stays constant, the signal current decreases and SNR usually becomes worse. Increasing bandwidth can improve speed but also increases integrated noise.

SNR can be improved by selecting suitable gain, using optical filters, reducing dark current, cooling when appropriate, optimizing bandwidth and improving optical coupling.

Design Decision 2: I recommend a low-noise detector, optical filtering and optimized receiver bandwidth. APD gain should be kept at a suitable level because excessive gain increases avalanche noise.

Stage 4 – Response Time and Bandwidth (10 Marks)

Given transit time tr = 20 ps = 20×10^{-12} s; approximate bandwidth $f \approx 0.44 / tr$

$f \approx 0.44 / (20 \times 10^{-12}) = 2.2 \times 10^{10}$ Hz

Answer: Approximate transit-time bandwidth = 22 GHz.

Carrier transit time is the time taken by generated carriers to cross the detector. A shorter transit time supports higher bandwidth. The RC time constant also limits speed: a larger resistance or capacitance increases RC delay and reduces bandwidth. A larger detector area improves optical collection but generally increases capacitance. Therefore, detector area, coupling, sensitivity and speed must be balanced.

Design Decision 3: I recommend a fast APD or suitable high-speed PIN receiver with low capacitance and optimized readout circuitry for LiDAR. The 22 GHz value is the transit-time estimate only; actual system bandwidth may be lower due to RC and circuit limits.

Stage 5 – PV, Image Sensors and Displays (15 Marks)

A. Solar photovoltaic analysis

Given: Voc = 0.60 V, Isc = 5 A, fill factor FF = 0.75

Maximum power Pmax = Voc × Isc × FF

Pmax = 0.60 × 5 × 0.75 = 2.25 W

Answer: Maximum PV power = 2.25 W, based on the given values.

Voc is the voltage at zero output current. Isc is the current when output voltage is zero. Fill factor measures how close the I–V curve is to an ideal rectangular shape. PV efficiency is Pmax divided by incident optical power. Flexible PV devices are lightweight and can bend to fit wearables, but may have lower efficiency, durability concerns and packaging challenges.

B. CMOS and CCD comparison

Parameter	CMOS	CCD
Architecture	Amplification/readout often integrated at pixel or column level	Charge is transferred across the chip to a readout node
Power	Usually lower	Usually higher
Speed	Fast readout and easy integration	Can provide high image quality but readout may be slower
Integration	Well suited to SoC integration	Less direct integration with processing circuits
Pixel scaling	Well suited to compact, low-power sensors	Scaling and readout complexity can be challenging
Applications	Face recognition, phones, automotive cameras	Scientific and specialized imaging

Smaller pixels allow more pixels in the same area, but collect fewer photons per pixel and can reduce low-light performance. Pixel scaling must balance resolution, noise, dynamic range and optical quality.

C. Emerging display comparison

Feature	OLED	MicroLED	Quantum-dot display
Light generation	Organic materials emit light	Microscopic inorganic LEDs emit light	Quantum dots convert light to selected colours; many designs use a backlight
Brightness	High	Very high potential	High, depending on design
Contrast	Excellent; pixels can turn off	Excellent	Good to excellent depending on backlight architecture
Efficiency/lifetime	Good, but organic materials can age	High potential efficiency and long lifetime	Good colour efficiency; depends on display architecture
Flexibility	Suitable for foldable displays	MicroLED transfer and flexible integration are challenging	Usually depends on the underlying panel/backlight
AR/VR/MR	Compact, high contrast	Promising for very bright compact displays	Good colour performance

Design Decision 4: I recommend flexible PV for wearable auxiliary power, a CMOS image sensor for face recognition and automotive integration, and OLED for compact foldable displays. MicroLED is a strong option where very high brightness and long lifetime are more important than cost and manufacturing complexity.

Stage 6 – Biomedical and Industrial Applications (10 Marks)

1. Wearable PPG

An LED illuminates the skin and a photodiode measures reflected light. Blood-volume changes alter the received light, producing a small AC component. A PIN photodiode with suitable responsivity, low-noise amplification, filtering and signal processing is suitable. Low power consumption and stable skin contact are important.

2. NIR spectroscopy

Near-infrared light is directed through or reflected from biological material. Different substances absorb and scatter light differently. A detector with good NIR responsivity, suitable optical filters and signal processing measures the returned intensity. Calibration is needed before estimating material properties.

3. Automotive LiDAR

LiDAR sends optical pulses and measures the returning light. The time delay is used to estimate distance. A sensitive APD or other suitable fast detector, high bandwidth, narrow optical filtering and low-noise receiver improve ranging performance.

4. Face recognition

A CMOS camera captures facial images. NIR illumination can help in low-light or controlled illumination conditions. Resolution, pixel sensitivity, image processing speed and power consumption affect performance.

5. UV sensing

UV sensing requires a detector responsive to the required ultraviolet wavelength range and with low response to unwanted visible or infrared light. Optical filters, calibration and suitable packaging improve reliable UV measurement.

Design Decision 5: I recommend PIN-based PPG sensing, an NIR-sensitive detector for spectroscopy, an APD-based high-speed LiDAR receiver, a CMOS camera for face recognition and a UV-optimized photodetector for ultraviolet sensing.

Stage 7 – MATLAB/SCILAB Simulation and Verification (5 Marks)

The following MATLAB/SCILAB-style code plots photocurrent versus optical power for a detector responsivity of 0.60 A/W and calculates the given bandwidth and SNR. This is a simple theoretical model; detailed noise simulation requires detector dark current, temperature, bandwidth and APD excess-noise parameters.

```
% Responsivity and bandwidth model
clear; clc; close all;
R = 0.60;                  % responsivity in A/W
P_uW = 0:5:100;            % optical power in micro-watts
P_W = P_uW * 1e-6;         % convert to watts
I_A = R .* P_W;            % photocurrent in amperes

plot(P_uW, I_A*1e6, 'o-');
xlabel('Optical Power (micro-W)');
ylabel('Photocurrent (micro-A)');
title('Photocurrent vs Optical Power');
grid on;

% Given calculations
deltaP = 25e-6;
I_PPG = 0.60 * deltaP;
I_LiDAR = 0.80 * 3e-6;
tr = 20e-12;
f = 0.44 / tr;
SNR_dB = -22 - (-37);
SNR_linear = 10^(SNR_dB/10);
Pmax = 0.60 * 5 * 0.75;

fprintf('PPG current variation = %.2f micro-A\n', I_PPG*1e6);
fprintf('LiDAR photocurrent = %.2f micro-A\n', I_LiDAR*1e6);
fprintf('Transit-time bandwidth = %.2f GHz\n', f/1e9);
fprintf('SNR = %.2f dB, linear ratio = %.2f\n', SNR_dB, SNR_linear);
fprintf('PV maximum power = %.2f W\n', Pmax);
```

Expected calculated outputs: PPG current variation = 15 μA; LiDAR photocurrent = 2.4 μA; transit-time bandwidth = 22 GHz; SNR = 15 dB (linear ratio 31.62); PV maximum power = 2.25 W.

Interpretation: The photocurrent plot should be linear because $I = R \times P$. The bandwidth is the theoretical transit-time limit and may be reduced by detector capacitance and receiver circuitry. The noise model should be extended with actual device parameters for a complete noise simulation.

Final System Verification

Requirement	Calculated / selected result	Status

PPG photocurrent variation	15 μA	Meets calculation
LiDAR photocurrent	2.4 μA	Meets calculation
Responsivity	0.60 A/W for PPG; 0.80 A/W for LiDAR	Given values used
Quantum efficiency	Not numerically calculable without wavelength	Needs wavelength
NEP / detectivity	Definitions given; no numerical detector data supplied	Needs detector data
Low-light SNR	15 dB; linear ratio 31.62	Calculated
Noise performance	Filtering, low dark current, optimized bandwidth and suitable gain	Design recommendation
High-speed bandwidth	22 GHz transit-time estimate	RC limit also applies
PV maximum power	2.25 W	Calculated
Image sensor	CMOS	Selected
Display	OLED for foldable/compact use	Selected
PPG / NIR	PIN-based PPG; NIR-sensitive detector for spectroscopy	Selected
LiDAR / automotive	High-speed APD receiver; CMOS camera for vehicle imaging	Selected
Face recognition	CMOS image sensor with suitable illumination	Selected
UV sensing	UV-optimized detector and optical filtering	Selected
MATLAB/SCILAB	Theoretical photocurrent plot and calculations	Code provided

Note: Quantum efficiency, NEP and detectivity cannot be calculated numerically from the assignment data alone because wavelength, noise current, detector area and other detector-specific values are not supplied.

Final System Design – Integrated Optoelectronic Platform (10 Marks)

Proposed block diagram (text form):

Optical Sources (visible LED / NIR LED / LiDAR laser / UV source) → Target or Sample → Optical Filters and Lens → Application-Specific Detector (PIN / APD / CMOS image sensor / UV detector) → Low-Noise Amplifier and Signal Conditioning → ADC and Processing Unit → Output (PPG/NIR information, LiDAR distance, face recognition, UV level) → OLED Display. Flexible PV → Power Management → Auxiliary power supply.

Final parameter/design table:

Subsystem	**Selected configuration**	**Reason**
PPG	PIN photodiode, R = 0.60 A/W	Low noise and suitable sensitivity
LiDAR	APD, R = 0.80 A/W before multiplication	Internal gain for weak returns and fast detection
Noise control	Optical filter, low-dark-current detector, optimized bandwidth	Improves weak-signal reliability
Detector speed	20 ps transit time; approximately 22 GHz limit	Suitable theoretical speed; RC must be checked
PV	Flexible PV for wearable auxiliary supply	Lightweight integration
Imaging	CMOS	Low power and SoC integration

Display	OLED	High contrast and foldable form factor
Biomedical	PPG and NIR sensing	Non-invasive optical measurement
Automotive	APD LiDAR plus CMOS camera	Distance measurement and image capture
Additional sensing	CMOS/NIR face recognition and UV-optimized detector	Application-specific response

Final Engineering Decision

YES, I would recommend developing and testing this integrated optoelectronic platform as a prototype. The calculated PPG current variation is 15 μA, LiDAR photocurrent is 2.4 μA, SNR is 15 dB, transit-time bandwidth is approximately 22 GHz and PV maximum power is 2.25 W. PIN photodiodes are suitable for low-noise PPG, while APDs are suitable for weak LiDAR returns. CMOS offers good integration and low power, and OLED is suitable for compact foldable displays.

However, I would not deploy it as a finished product until detector NEP, detectivity, quantum efficiency, noise performance, optical safety, thermal behaviour, reliability and full MATLAB/SCILAB noise simulations are verified with real component specifications.

Conclusion

In this assignment, I analysed photodetectors, noise, SNR, detector bandwidth, photovoltaic power, image sensors, display technologies and application-specific sensing. The calculated values support the proposed design at a theoretical level. PIN and APD detectors, CMOS imaging, flexible PV and OLED display technologies provide a practical starting configuration. Further device-specific simulation and experimental testing are required before final deployment.

References

1. ECA18 – Optoelectronics and Optical Communication, CO3 Scenario-Based Assignment (provided assignment sheet).

2. Standard photodetector relations: responsivity, quantum efficiency, noise-equivalent power and detectivity, as covered in optoelectronics course materials.

3. MATLAB or GNU Octave documentation for plotting and numerical calculations.